

VOLUME 4 • B.TECH CSE STUDY SERIES

HTML & HTML5

Complete Professional Handwritten Notes

*Web Document Architecture, DOM Tree, Semantic Tags, Forms &
Input Validation, Media & Accessibility*

```
<!DOCTYPE html>
<main class="web">
  <article>Semantic</ar
</main>
```

✓ HTML Document Structure &
DOM Tree

✓ HTML5 Semantic Layout Tags

✓ Advanced Form Controls &
Validation

✓ Audio, Video, iFrames & Canvas

✓ SEO Meta Tags & Accessibility
(ally)

✓ 30+ Viva & Web Tech Interview
Q&A

Designed for University Web Tech Exams & Frontend Engineering Roles



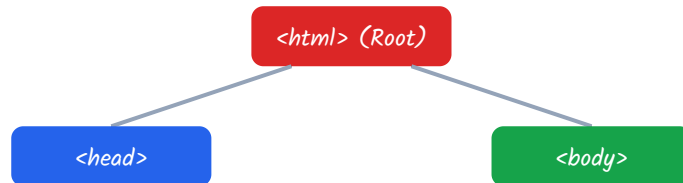
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CHAPTER 2: HTML DOCUMENT STRUCTURE & DOM TREE

💡 HTML5 Boilerplate & DOM Hierarchy

SVG DIAGRAM: HTML DOCUMENT OBJECT MODEL (DOM) TREE



```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSE Web Page</title>
</head>
<body>
  <header>
    <h1>Welcome to CSE Portal</h1>
  </header>
  <main>
    <article>
      <h2>HTML5 Semantics</h2>
      <p>Semantic markup improves SEO and accessibility.</p>
    </article>
  </main>
</body>
</html>
```

CHAPTER 6: SEMANTIC HTML5 ARCHITECTURE



Semantic vs Non-Semantic Tags

Tag	Type	Semantic Purpose
<header>	Semantic	Represents introductory content, site logo, or top navigation.
<nav>	Semantic	Defines a section containing navigation links.
<main>	Semantic	Encapsulates the dominant content unique to the document.
<article>	Semantic	Self-contained composition (blog post, news story, comment).
<section>	Semantic	Standalone thematic grouping of content with a heading.
<div>	Non-Semantic	Generic container for styling purposes without inherent meaning.

CHAPTER 12: 30+ VIVA & INTERVIEW QUESTIONS

Q1. What is the purpose of the alt attribute in img tags?

Ans: Provides alternative text for screen readers (accessibility) and displays if the image fails to load.

Q2. What is the difference between inline and block elements?

Ans: Block elements (e.g. `<div>`, `<p>`, `<h1>`) start on a new line and take full container width. Inline elements (e.g. `<span>`, `<a>`, `<b>`) do not start on a new line and take only as much width as necessary.